

Path integral control of partially observed systems via fully observable control approximations

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ABSTRACT

This study considers a partially observed optimal control problem in which the state is measured through a noisy output, focusing on continuous-time and continuous-space settings. To develop a control algorithm for this challenging problem, we extend the so-called fully observable value approximation, originally developed for the partially observable Markov decision problems (POMDPs) in discrete-time and discrete-space domains, to the continuous ones. The approximation yields a suboptimal controller with a structure analogous to that of a controller obtained through the separation principle for the so-called linear-quadratic settings. Combined with the path integral control method, the proposed extension allows for a suboptimal policy synthesis for partially observable systems. Numerical experiments demonstrate the effectiveness of the proposed algorithm in handling partially observable control problems of continuous-time, continuous-space, and nonlinear systems.

1. Introduction

Optimal control of partially observable stochastic systems has been recognized as a challenging problem since the early days of control theory [1]. In numerous practical examples, the complete measurement of the system state is often costly or technically impossible; solving the control problem through a noisy output is essential to handling practical control problems. The challenging problems necessitate control strategies that rely on the noisy observation rather than the exact measurement of a system state, differentiating them from fully observed scenarios. The efforts for solving the partially observed problems has been made for various settings including both continuous and discrete settings [2–7].

The partially observed control problems pose significant challenges in developing control algorithms. The problems under the discrete-time and discrete-space settings are generally addressed within the context of partially observed Markov decision processes (POMDPs) [1]. Even in the discrete problems, determining the optimal control is quite challenging. Solving the POMDPs generally reduces to determining parameters, the so-called alpha vectors, to obtain value functions in the dynamic programming approach [8]. Determining the alpha vectors is achieved by the enumeration of the candidate vectors, leading to computational complexity issues. From a computational complexity perspective, the studies [9,10] reveal that as the number of control states and inputs increases, such a strategy becomes computationally

intractable. Accordingly, several approximation techniques have been proposed to mitigate the difficulty. Such developments include the fully observable value approximation [6, Section 21.1] such as MDP and QMDP approximations [11,12] and the point-based value iteration algorithm [13,14].

Continuous-time and continuous-space problems introduce further challenges, except for the so-called linear-quadratic setting, for which the separation principle is applicable [15]. The partially observed control problem in general settings necessitates solving control and estimation problems simultaneously. Accordingly, the problem is formulated as the control problem of the so-called Zakai equation [16], which is a stochastic partial differential equation (SPDE) for the filtering of nonlinear systems. The SPDE presents significant challenges from both theoretical and numerical perspectives [2]. In particular, such a problem may be solved numerically through the discretization of the equations in both space and time, with the application of POMDP algorithms developed for discrete settings. However, this strategy tends to be computationally intractable, due to the aforementioned computational complexity in discrete POMDP algorithms. Another approach to the continuous problem is to employ Pontryagin's maximum principle-type conditions through the forward and backward stochastic differential equations (FBSDEs) [17,18]. It should be noted that in recent years, numerical methods have been developed by leveraging the maximum principle based approach [19,20]. Considering

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the development of the model predictive control for deterministic control problems [21], the development of a practical algorithm for the partially observed problems remains challenging.

This study presents a control method for the partially observed control of nonlinear systems in continuous-time and continuous-space settings. Specifically, to mitigate the difficulties in solving the control problems, we develop an approximation strategy that allows us to obtain suboptimal controllers which can be practically implemented. To this end, we employ an analogue to the fully observable value approximation, which has been originally developed for discrete POMDPs [11, 12]. This employs a strategy to approximately solve a control problem as if it were a fully observed control problem given a conditional probability of a state through an estimation. It has been demonstrated that the approximation yields a suboptimal controller for discrete POMDPs and exhibits satisfactory performance in numerous examples, although it is a straightforward and heuristic approach [12]. In light of these considerations, we have developed a counterpart of the fully observable value approximation for continuous-time and continuous-space problems. Extending the method to the continuous setting is not a straightforward process, as the approximation method in discrete scenarios heavily relies on discrete structures. Main contributions of this study are developing a continuous analogue of the fully observable value approximation and obtaining a suboptimal controller based on the approximation. It should be noted that the approximation strategy allows us to obtain an expression of the suboptimal controller. This is in contrast to the discrete setting of the fully observable value approximation, which only yields an approximate value function. The suboptimal controller has a form of a conditional expectation of the optimal controller for a fully observed optimal control problem. This form can be regarded as a generalization of the optimal controller for the partially observed optimal control problems under the so-called linear-quadratic setting, obtained through the separation principle.

Another contribution of this study is developing a control algorithm by integrating the controller obtained through the approximation with a filtering method in the continuous settings. Implementing the controller obtained through the fully observable approximation necessitates solving fully observed optimal control problems for nonlinear stochastic systems, which is computationally challenging in general. To address this challenge, we employ the path integral control framework [22–25], which enables us to solve the fully observed optimal control problem through a sampling-based approach. Furthermore, in the context of partially observed control problems, filtering is a crucial step before determining the control to estimate the system state. This study integrates a particle filtering algorithm called the path integral particle filtering (PIPF) algorithm, recently developed in [26,27], with the control algorithm. The PIPF algorithm employs the path integral to address the filtering problem, thereby enabling a seamless integration of the control and filtering algorithms through the path integral framework. The path integral allows the control problem to be solved through sampling; our method effectively handles nonlinear control problems and provide sufficient performance, as demonstrated through numerical experiments in this paper.

This paper is organized as follows. The following section introduces the fully observed approximation for discrete POMDPs. Subsequently, the partially observed control problems are formulated in a continuous-space and continuous-time setting in Section 3, and the fully observable approximation for continuous problems is discussed in Section 4. After presenting the suboptimal controller obtained through the approximation, we present the control algorithm employing the path integral control in Section 5. Finally, we demonstrate the performance of the proposed control method through numerical examples in Section 6.

Notations. The set of the real numbers is denoted by $\mathbb{R}$, and $\mathbb{R}^d$ represents the d -dimensional Euclidean space. $\mathbb{R}^{n \times m}$ denotes the space of real-valued matrices of dimension $n \times m$. The d -dimensional identity matrix is denoted by I_d . The trace of a matrix M is denoted by $\text{Tr}(M)$. For a finite set A , the number of its elements is denoted by $|A|$. Given

a random variable X which follows a probability distribution π , we write the relation by $X \sim \pi$. We usually represent a probability space by $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is the sample space, $\mathcal{F}$ is the σ -algebra on Ω , and $\mathbb{P}$ is the probability measure. Given the probability space, the probability of an event $A \in \mathcal{F}$ is denoted by $\mathbb{P}(A)$, and the conditional probability of the event A under an event B is denoted by $\mathbb{P}(A | B)$. The expectation of a random variable X is denoted by $\mathbb{E}[X]$, and the conditional expectation of X under an event B or a sub σ -algebra $\mathcal{G} \subset \mathcal{F}$ are denoted by $\mathbb{E}[X | B]$ and $\mathbb{E}[X | \mathcal{G}]$, respectively. Additionally, given a random variable X , the σ -algebra generated by the random variable X is denoted by $\sigma(X)$.

2. Preliminaries: Fully observable value approximation algorithm for POMDPs

2.1. POMDPs in discrete settings

We provide a brief overview of the fully observable value approximation algorithm for POMDPs in discrete-time and discrete-space settings, which serves as a foundation for developing the continuous-time algorithm. First, we present a formulation of a discrete POMDP, followed by the overview of the fully observable value approximation. The problem presented below is a discrete counterpart of the continuous-time partially observable control problem formulated in Section 3. The presentation of the discrete POMDP formulation follows [28, Chapter 9].

A standard POMDP considers finite sets of state, control, and observation variables, denoted by $\mathcal{X} = \{x^1, \dots, x^{|\mathcal{X}|}\}$, $\mathcal{U} = \{u^1, \dots, u^{|\mathcal{U}|}\}$, and $\mathcal{Y} = \{y^1, \dots, y^{|\mathcal{Y}|}\}$, respectively. The control problem assumes that we are given a dynamical system governing the transition of the $\mathcal{X}$ -valued state X_t ($t = 0, \dots, T$) and that the $\mathcal{Y}$ -valued observation Y_t ($t = 0, \dots, T$) are obtained through a noisy measurement of the state X_t . The transition probability of the state, $p_T : \mathcal{X} \times \mathcal{U} \times \mathcal{X} \rightarrow [0, 1]$, and the observation probability $p_O : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$ are defined as follows:

$$p_T(x' | x, u) = \mathbb{P}(X_{t+1} = x' | X_t = x, u_t = u) \quad (1)$$

$$p_O(y | x) = \mathbb{P}(Y_t = y | X_t = x) \quad (2)$$

for $x, x' \in \mathcal{X}$, $u \in \mathcal{U}$, and $y \in \mathcal{Y}$.

We consider the conditional probability of X_t given the history of Y_t . More precisely, this is considered as the conditional probability of X_t given the sub σ -algebra $\mathcal{F}_t^Y = \sigma(Y_0, \dots, Y_t)$, which is usually referred to as the belief in discrete problem settings. In the following, the belief at time t is denoted by b_t , which is defined as

$$b_t(x) = \mathbb{P}(X_t = x | \mathcal{F}_t^Y) \text{ for } x \in \mathcal{X}. \quad (3)$$

It is known that the belief is a sufficient statistic, and its dynamics of the belief vector $b_t = (b_t(x^1), \dots, b_t(x^{|\mathcal{X}|}))$ are governed by the equation derived from Bayes' formula for the filtering problem,

$$b_{t+1} = \hat{F}(b_t, u_t, Y_{t+1}) \quad (4)$$

where $\hat{F}$ is given by

$$\hat{F}(b, u, y')(x') = \frac{p_O(y' | x') \sum_{x \in \mathcal{X}} p_T(x' | x, u) b(x)}{\sum_{x' \in \mathcal{X}} p_O(y' | x') \sum_{x \in \mathcal{X}} p_T(x' | x, u) b(x)}. \quad (5)$$

for each $x' \in \mathcal{X}$.

The POMDP is formulated as a Markov decision problem regarding the belief dynamics (4). A standard form of the cost function for a control policy ϕ is given by

$$J_t(\phi, b) = \mathbb{E}_\phi \left[\sum_{\tau=t}^T \hat{\ell}_\tau(b_\tau, u_\tau) | b_t = b \right] \quad (6)$$

where the mathematical expectation is taken with respect to the path of the belief under the policy ϕ and $\hat{\ell}_\tau(b, u)$ is the stage cost function usually given in the form of the conditional expectation of a function,

$$\hat{\ell}_\tau(b_\tau, u_\tau) = \sum_{x \in \mathcal{X}} \ell_\tau(x, u_\tau) b_\tau(x) \quad (7)$$

with a function $\ell_\tau : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$ for $\tau = 0, 1, \dots, T$. The problem is formulated as finding the optimal policy that minimizes the cost function (6) given the belief dynamics (4).

The value function of the POMDP is represented by

$$V_t(b) = \mathbb{E}_{\phi^*} \left[\sum_{\tau=t}^T \hat{\ell}_\tau(b_\tau, u_\tau) \mid b_t = b \right] \quad (8)$$

where ϕ^* represents the optimal policy of the POMDP. The Bellman principle implies that if we know the value function, it follows the Bellman backward equation

$$V_t(b) = \min_{u \in \mathcal{U}} \left[\hat{\ell}_t(b, u) + \mathbb{E}_u [V_{t+1}(b') \mid \mathcal{F}_t^Y] \right] \quad (9)$$

where $b' = \hat{F}(b, u, Y_{t+1})$ is determined by the belief dynamics (4). Eq. (9) may allow us to solve the POMDP. However, the problem generally poses a significant challenge in determining the value function due to the aforementioned computational complexity issues [9,10]. This motivates the developments of approximation strategies in the discrete POMDPs, including the fully observable value approximation introduced below.

2.2. Fully observable value approximation for POMDPs

We briefly introduce the fully observable value approximation as a preliminary to develop a continuous counterpart for the continuous-time partially observed control problems. We present a finite-horizon POMDP (see [8] for the finite-time setting of POMDPs). To present the approximation with the finite horizon, we employ slight modifications to an infinite horizon framework that is typically discussed in existing literature [12]. The fully observable value approximation utilizes the value function of a fully observed Markov decision problem. The fully observed control problem is formulated as the minimization problem of the cost

$$J_t^F(\phi^F, x) = \mathbb{E}_{\phi^F} \left[\sum_{\tau=t}^T \ell_\tau(X_\tau, u_\tau) \mid X_t = x \right] \quad (10)$$

where ϕ^F is a control policy for the fully observed control problem which assumes the complete observation of the state X_t governed by the transition probability p_T of (1). The value function of the fully observed problem is defined by

$$V_t^F(x) = \mathbb{E}_{\phi^{F,*}} \left[\sum_{\tau=t}^T \ell_\tau(X_\tau, u_\tau) \mid X_t = x \right] \quad (11)$$

where the expectation is taken over the path $\{X_\tau\}_{\tau=t}^T$ given $X_t = x$ using the optimal policy $\phi^{F,*}$ for the fully observed control problem.

The fully observable value approximation computes the approximate value function of the POMDP (see [12, Section 4.1] for the infinite horizon setting of the approximation). The method introduces the approximate value function of the POMDP at time t , denoted by V_t^A , by

$$V_t^A(b) = \min_{u \in \mathcal{U}} \left[\hat{\ell}_t(b, u) + \sum_{x', x'' \in \mathcal{X}} p_T(x' \mid x, u) b(x'') V_{t+1}^F(x') \right], \quad (12)$$

where A represents the approximation. This approximation is usually referred to as the QMDP approximation [12, 4.1.2]. This equation is obtained by modifying the Bellman equation (9). Intuitively, this approximation disregards the partially observable setting after time t to approximate the value function and the control of the POMDP at that time. In other words, this strategy determines the control by considering only the system transition dynamics, assuming that the exact state value is available after t , given the conditional distribution given the observation history up to that time. A suboptimal control at time t is obtained as the minimizer $u^* \in \mathcal{U}$ of the right-hand side of (12).

The approximation strategy often shows sufficient performance in numerical experiments [12], although it employs a straightforward approximation. This study develops a continuous-time counterpart to this fully observable value approximation to develop a practical algorithm for the continuous partially observed control problem formulated in the next section.

3. Problem formulation: Partially observed optimal control problem in continuous setting

This section presents the problem formulation of the partially observed control problem addressed in this study. We address the problem of nonlinear systems in continuous time and continuous space.

The state and observation equations are described by the stochastic differential equations (SDEs)

$$dX_t = f(X_t)dt + \sigma(X_t)(u_t dt + \sqrt{\lambda} dW_t), \quad X_0 \sim \pi_0 \quad (13a)$$

$$dY_t = h(X_t)dt + \sigma_B dB_t, \quad Y_0 = 0, \quad (13b)$$

where X_t is the $\mathbb{R}^n$ -valued state process, u_t is the $\mathbb{R}^m$ -valued control process, Y_t is the $\mathbb{R}^p$ -valued observation process, W_t and B_t are mutually independent standard Wiener processes of the dimensions of m and p , respectively. Additionally, $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$ are locally Lipschitz continuous functions, $\lambda, \sigma_B > 0$ are constants, and π_0 is the probability distribution of the initial state X_0 . The control input and the Wiener process W_t share the coefficient $\sigma(X_t)$ in (13a), which is a standard assumption in the path integral control framework employed in this study.

This study is interested in the partially observed control problem of system (13). This means the control u_t is determined based on the history of noisy observations $\mathcal{Y}_t = \{Y_s\}_{0 \leq s \leq t}$, rather than the system state. In other words, the control u_t is assumed to be an $\mathcal{F}_t^Y$ -adapted process where $\mathcal{F}_t^Y = \sigma\{Y_s, 0 \leq s \leq t\}$ is the σ -algebra generated by the observation Y_t . To formulate the optimal control problem, we introduce the cost functional

$$J(t, u) = \mathbb{E} \left[\int_t^T L(X_s, u_s) ds + \Phi(X_T) \mid \mathcal{F}_t^Y \right] \quad (14)$$

where $u \in \mathcal{A}_{[t,T]}$ with $\mathcal{A}_{[t,T]}$ being the admissible set of the control process on the time horizon $[t, T]$ and the stage cost $L : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is given by

$$L(x, u) = \ell(x) + \frac{1}{2} u^T u \quad (15)$$

with the function $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$. We employ the form of L given by (15) to apply the path integral control algorithm later. Function $\Phi : \mathbb{R}^n \rightarrow \mathbb{R}$ is the terminal cost. As in standard optimal control problems, we assume ℓ and Φ are non-negative valued functions for simplicity.

The partially observed control problem is formulated as follows:

Problem 1. Consider the state and observation equations in (13) and the cost functional (14). Find a control that minimizes cost functional (14), given the history of the control and the observation.

Note that this control problem is more challenging than the fully observed control problem; solving this problem has to account for the fact that the state is not directly accessible. Furthermore, the control has to be determined based on the unreliable observations, which are outputs of the system perturbed by noise. The partially observed control problem is usually reformulated as the control problem of conditional distributions or densities, governed by SPDEs, often referred to as the Zakai equation. Specifically, in the current setting, the control problem is formulated as the minimization of the cost given by

$$J(t, u, q) = \mathbb{E} \left[\int_t^T \langle L(\cdot, u_s), q_s \rangle ds + \langle \Phi(\cdot), q_T \rangle \mid q_t = q \right] \quad (16)$$

where $\mathbb{E}$ denotes the mathematical expectation with respect to the probability measure $\mathbb{P}$, which is induced by a change of measure and differs

from the original probability measure $\mathbb{P}$. Moreover, $\langle L(\cdot, u_s), q_s \rangle = \int_{\mathbb{R}^n} L(x, u_s) q_s(x) dx$, and q_t is the unnormalized conditional probability density function governed by the Zakai SPDE:

$$dq_t = A_{u_t} q_t dt + \sum_{k=1}^p S^k q_t dY_{k,t} \quad (17)$$

with the operators A_{u_t} and S^k determined by the coefficients of the state and observation equations in (13) (see [29, 7.6] or [30, Example 6.1] for the details). The Zakai equation is known as the filtering equation, yielding unnormalized conditional probability distributions of the state given the history of a noisy observation, for continuous-time systems composed of the state and observation Eqs. (13). Note that cost (16) and Eq. (17) correspond to cost (6) and the belief dynamics (4) in the discrete case, respectively.

Solving the partially observed control problems is theoretically and numerically challenging. In particular, the SPDE can be handled numerically by the discretization. However, it becomes computationally intractable due to the curse of dimensionality. The application of discrete POMDP algorithms to the discretized control problem becomes quite challenging due to the aforementioned computational complexity issue. Motivated by the discussion, this study focuses on developing a practical control method for solving the partially observed control problems by developing a continuous analogue of the fully observable approximation.

4. Fully observable control approximation for continuous problems

Building on the fully observable value approximation for the discrete POMDPs, we develop a continuous analogue for the partially observable control problems formulated in Section 3.

The proposed framework assumes the conditional distribution of the state X_t of system (13a) given the observation history $\mathcal{Y}_t = \{Y_s\}_{0 \leq s \leq t}$ through the observation equation (13b). As seen from the discussion on the Zakai equation, the conditional distribution of the state X_t given the observation $\mathcal{Y}_t$ is a measure-valued stochastic process, denoted by π_t^Y , such that it holds that

$$\mathbb{E}[\phi(X_t) | \mathcal{F}_t^Y] = \int_{\mathbb{R}^n} \phi(x) d\pi_t^Y(x). \quad (18)$$

for any bounded measurable function $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$, as shown in [31]. Note that π_t^Y is regarded as the continuous counterpart of the belief for the discrete POMDPs, discussed in Section 2. The conditional distribution π_t^Y can be obtained through the normalization of the unnormalized conditional density function q_t which is the solution to the Zakai equation (17). Instead of the SPDE approach, this study employs particle approximation in the filtering to estimate the conditional distribution. More precisely, we consider particles $\{\tilde{X}_t^k\}_{k=1, \dots, K}$ and the associated nonnegative weights $\{w_f^k\}_{k=1, \dots, K}$ such that $\sum_{k=1}^K w_f^k = 1$ and

$$\pi_t^Y \approx \sum_{k=1}^K w_f^k \delta_{\tilde{X}_t^k} \quad (19)$$

where $\delta_{\tilde{X}_t^k}$ is the Dirac delta at the point $\tilde{X}_t^k$. We consider the approximation in (19) holds in the sense that the particle approximation in the right-hand side converges to π_t^Y in the weak topology as the number K of particles tends to the infinity [31].

We develop the continuous counterpart of the fully observable approximation upon the discussion on the discrete POMDP in Section 2. Given the conditional distribution π_t^Y at time $t \in [0, T]$, we consider the continuous analogue of the fully observable approximation of the value function in (12) by

$$V^A(t, \pi_t^Y) = \min_{u \in \mathcal{A}_{[t, T]}^A} \mathbb{E} \left[\int_t^T L(X_s, u_s) ds + V^F(t_1, X_{t_1}) | \mathcal{F}_t^Y \right], \quad (20)$$

where $0 \leq t \leq t_1 \leq T$ and $\mathcal{A}_{[t, T]}^A$ is a class of the admissible control set of $\mathbb{R}^m$ -valued continuous processes u_s for $s \in [t, T]$ adapted to the

filtration composed by $\mathcal{F}_t^Y$ and the filtration generated by X_s , which is the solution to (13a), for $s \in [t, T]$. Note that the control algorithm developed below considers the approximation $V^A(t, \pi_t^Y)$ at each time step t on $[0, T]$ and applies u_t , chosen from $u \in \mathcal{A}_{[t, T]}^A$ to minimize (20), to system (13) at each t , as in the model predictive control framework. The control u_t , applied to the actual system in this manner, is adapted to $\mathcal{F}_t^Y$. The function $V^F : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ is the value function of the fully observed control problem under the state equation (13a), defined by

$$V^F(t_1, x) = \min_{u \in \mathcal{A}_{[t_1, T]}^F} \mathbb{E} \left[\int_{t_1}^T L(X_s, u_s) ds + \Phi(X_T) | X_{t_1} = x \right], \quad (21)$$

where $\mathcal{A}_{[t_1, T]}^F$ is the admissible set of the control process adapted to the filtration generated by the state process X_s for $s \in [t_1, T]$ given $X_{t_1} = x$. Note that the mathematical expectations in (20) and (21) are taken with respect to the path of X_t governed by the state equation (13a) under the control u_t . The approximate value function $V^A(t, \pi_t^Y)$ is a counterpart of (12) in the sense that given the conditional distribution π_t^Y of X_t at time t , we consider only the state transition in (20) after that time to determine the control. In other words, we ignore the observation equation (13b) after the time instant in (20).

As for (20), using the properties of (18) and (19), the particle approximation of the conditional distribution allows the approximation of $V^A(t, \pi_t^Y)$ by the approximate value function $\tilde{V}^A(t, \pi_t^Y)$ given by

$$\tilde{V}^A(t, \pi_t^Y) = \min_{u \in \mathcal{A}_{[t, T]}^A} \sum_{k=1}^K w_f^k \mathbb{E} \left[\int_t^{t_1} L(X_s^k, u_s) ds + V^F(t_1, X_{t_1}^k) \right] \quad (22)$$

for $0 \leq t \leq t_1 \leq T$, where X_s^k is a solution of the system equation (13a) over $[t, t_1]$ starting from $X_t^k = \tilde{X}_t^k$ being a particle in the particle approximation of the conditional distribution π_t^Y .

This approximation yields the minimizer of (22). Assuming that $V^F(t_1, x)$ is twice continuously differentiable with respect to x and applying the Ito formula, we obtain

$$\begin{aligned} \tilde{V}^A(t, \pi_t^Y) &= \min_{u \in \mathcal{A}_{[t, T]}^A} \sum_{k=1}^K w_f^k \mathbb{E} \left[\int_t^{t_1} L(X_s^k, u_s) ds \right. \\ &\quad \left. + V^F(t_1, X_{t_1}^k) - V^F(t, X_t^k) + V^F(t, X_t^k) \right] \\ &= \min_{u \in \mathcal{A}_{[t, T]}^A} \mathbb{E} \left[\int_t^{t_1} \sum_{k=1}^K w_f^k \{ L(X_s^k, u_s) + \mathcal{L}V^F(s, X_s^k) \} ds \right. \\ &\quad \left. + \sum_{k=1}^K w_f^k V^F(t, X_t^k) \right], \end{aligned} \quad (23)$$

where $\mathcal{L}$ is the infinitesimal generator of the state equation (13a) defined by

$$\mathcal{L}V^F(t, x) = \frac{\partial V^F}{\partial t} + \frac{\partial V^F}{\partial x} (f(x) + \sigma(x)u) + \frac{1}{2} \text{Tr} \left(\sigma(x)^T \frac{\partial^2 V^F}{\partial x^2} \sigma(x) \right) \quad (24)$$

Note that we omit the arguments of the derivatives of V for notational brevity in (24). Furthermore, recalling the definitions of $L(x, u)$ in (15) and $\mathcal{L}V^F(t, x)$ in (24), completing the square with respect to the control variable shows that for any $u_s \in \mathbb{R}^m$,

$$\begin{aligned} &\sum_{k=1}^K w_f^k (L(X_s^k, u_s) + \mathcal{L}V^F(s, X_s^k)) \\ &\geq \sum_{k=1}^K w_f^k \ell(X_s^k) - \frac{1}{2} \left\| \sum_{k=1}^K w_f^k \frac{\partial V^F}{\partial x}(s, X_s^k) \sigma(X_s^k) \right\|^2 \\ &\quad + \sum_{k=1}^K w_f^k \left(\frac{\partial V^F}{\partial t} + \frac{\partial V^F}{\partial x} f(X_s^k) + \frac{1}{2} \text{Tr} \left\{ \sigma(X_s^k)^T \frac{\partial^2 V^F}{\partial x^2} \sigma(X_s^k) \right\} \right). \end{aligned} \quad (25)$$

holds with probability one. The equality holds when u_s is chosen as

$$u_s^* = - \sum_{k=1}^K w_f^k \sigma(X_s^k)^T \frac{\partial V^F}{\partial x}(s, X_s^k), \quad s \in [t, t_1]. \quad (26)$$

That is, the minimum in (23) is achieved by u_s^* given by (26) for $s \in [t, t_1]$. The above discussion yields the following result.

Proposition 1. Assume the following: (1) at time $t \in [0, T]$, we are given the particles $\{\tilde{X}_t^k\}_{k=1,\dots,K}$ and the associated weights $\{w_t^k\}_{k=1,\dots,K}$ in the particle approximation of the conditional distribution π_t^Y , (2) $\{X_s^k\}_{k=1,\dots,K}$ are solutions to the SDE (13a) for $s \in [t, t_1]$ with $X_t^k = \tilde{X}_t^k$, where $0 \leq t \leq t_1 \leq T$, and (3) the function $V^F(s, x)$ given by (21) is twice continuously differentiable with respect to x on $\mathbb{R}^n$ for $s \in [t, t_1]$. Then, the controller (26) achieves the minimum of the right-hand side of (22).

Note that the controller (26) is obtained as a consequence of introducing the continuous analogue of the fully observable approximation for the discrete POMDP. This study employs this controller as a suboptimal controller for solving the partially observed control problem in the continuous setting, formulated in Section 3. Notably, this result is useful in developing the numerical algorithm because it allows to determine the control at each time $t \in [0, T]$ by using the formula (26) if we know the function $V^F(t, \cdot)$ and the particle approximation of the conditional distribution π_t^Y for each t .

This controller (26) has a form of an empirical conditional mean of the optimal controller $-\sigma(x)^T \frac{\partial V^F}{\partial x}(t, x)^T$ for a fully observed control problem formulated under the state equation (13a) and the cost function (14). The suboptimal controller (26) coincides with the optimal controller designed by the separation principle for linear systems under the so-called linear-quadratic settings [15]. Indeed, the optimal control in the linear-quadratic settings adopts the form $u_t = K_t \mathbb{E}[X_t | \mathcal{F}_t^Y]$, where K_t is a time-varying feedback gain matrix and $\mathbb{E}[X_t | \mathcal{F}_t^Y]$ is the conditional mean of the state given the observation, typically computed using the Kalman filter. This controller is a modification of the optimal control for the fully observed control problem, which is $u_t = K_t X_t$ using the same gain matrix (see [32, Chapters 8 and 9] on the relation between optimal controls in fully and partially observed control problems of linear quadratic settings, for example). Thus, the controller (26) can be regarded as a generalization of the optimal control for the well-known case.

The implementation of the controller (26) requires computing the value function V^F for the fully observable control problem as well as the conditional distribution. Computing V^F is challenging even though it is the value function for the fully observed control problem, which is generally easier to solve than the partially observed one. This study overcomes this challenge by using the path integral control algorithm as described in the following section.

Remark 1. The theoretical analysis of the proposed algorithm is an important and interesting direction for future research. Specifically, it involves examining the performance gap between the true optimal controller for the partially observed control problem and the suboptimal controller obtained using the proposed algorithm. For discrete POMDPs, several studies have analyzed the performance of suboptimal policies obtained through POMDP algorithms in terms of cost functions [12,33]. Extending such performance analysis to the continuous setting requires an analysis based on the Zakai equation (17), which describes the unnormalized conditional probability density function of the state. It is known that the Zakai equation (17) reduces to the Fokker–Planck equation, which describes the probability density function of the state as determined by the state equation (13a) and corresponds to the fully observable setting, when the stochastic term in (17) is ignored. In other words, the fully observable approximation can be analyzed through the Fokker–Planck equation, which shares structural similarities with the Zakai equation. The performance gap between the true optimal controller and the fully observable approximate controller can be analyzed through analysis of these PDEs, leveraging functional-analytical tools. This topic is left for future work.

5. Algorithm for partially observed control problem

5.1. Brief review of path integral control

We present a control algorithm to address the partially observed control problem for system (13) based on the suboptimal controller (26). As mentioned in the end of the previous section, developing the algorithm necessitates the optimal controller for the fully observed optimal control problem. The key ingredient to solve this problem is the path integral control algorithm [22,24,34] (see also [35] for recent developments). We briefly introduce the path integral control in the beginning of this section, followed by the proposed algorithm.

The path integral control [22,24,34] addresses the fully observed optimal control problem of system (13a), that is, the minimization of cost functional (14) subject to state equation (13a) with respect to the control under the complete observation of the state X_t .

The path integral control yields the optimal control for system (13a) by utilizing path integrals, avoiding directly solving the nonlinear Hamilton–Jacobi–Bellman (HJB) equations arising in the standard approach of the optimal control problems. For obtaining optimal control of the problem, the path integral framework employs the following PDE

$$\frac{\partial \psi}{\partial t} = \frac{1}{\lambda} \mathcal{L} \psi - \frac{\partial \psi}{\partial x} f - \frac{\lambda}{2} \text{Tr} \left\{ \frac{\partial^2 \psi}{\partial x^2} \sigma^T \sigma \right\}, \quad (27)$$

with the boundary condition $\psi(T, x) = \exp(-\Phi(x)/\lambda)$ and the controller $u^* = \lambda \psi(t, x)^{-1} \sigma(x)^T \frac{\partial \psi}{\partial x}(t, x)^T$. Given the solution $\psi(t, x)$ to PDE (27), a change of variable $V(t, x) = -\lambda \log(\psi(t, x))$ recovers the HJB equation associated with the fully observed optimal control problem from PDE (27). That is, the function V coincides with V^F of (21). Additionally, in terms of function V , u^* can be expressed as $u^* = -\sigma(x)^T \frac{\partial V}{\partial x}(t, x)^T$, and it is proven to be optimal [22].

A notable feature of this framework is that it provides a solution to the linear PDE (27) in the form of a path integral expression, using the Feynman–Kac formula [36, Theorem 7.6], which leads to a sampling-based approach for optimal control problems. As shown in [22], the Feynman–Kac formula and the Girsanov theorem [36, Theorem 5.1] indicate that the function $\psi(t, x)$ given by

$$\psi(t, x) = \mathbb{E}_x^u \left[\exp \left(-\frac{1}{\lambda} S^u(t) \right) \right] \quad (28)$$

solves PDE (27), where

$$S^u(t) = \int_t^T \left(\mathcal{L}(X_s) + \frac{1}{2} u_s^T u_s \right) ds + \sqrt{\lambda} \int_t^T u_s^T dW_s + \Phi(X_T) \quad (29)$$

and $\mathbb{E}_x^u$ denotes the conditional expectation with respect to the path $\{X_s\}_{t \leq s \leq T}$ of SDE (13a) with an arbitrary control process u_t that is adapted to X_t given the condition $X_t = x$.

The expression (28) facilitates the computation of the value function through the previously mentioned change of variables. The study [34] further provides an expression of the optimal control particularly suitable for sampling-based algorithms. The following is taken from [34] with a minor modification.

Theorem 1. For any given control $u(t, x)$, the optimal controller $u^*(t, x)$ for the fully observed optimal control problem is characterized by

$$u^*(t, x) = u(t, x) + \lim_{\tau \rightarrow t} \frac{\mathbb{E}_x^u \left[\exp \left(-\frac{1}{\lambda} S^u(t) \right) \int_t^\tau \sqrt{\lambda} dW_s \right]}{(\tau - t) \mathbb{E}_x^u \left[\exp \left(-\frac{1}{\lambda} S^u(t) \right) \right]} \quad (30)$$

for each (t, x) , where $S^u(t)$ is given by (29) and $\mathbb{E}_x^u$ is the mathematical expectation with respect to the paths of SDE (13a) starting from $X_t = x$ given the control $u(t, x)$.

Path integral control (30) enables the implementation of the optimal control by sampling paths of SDE (13a) using a suboptimal control $u(t, x)$ designed a priori. To approximate the second term of (30), we sample paths from a time discretized SDE (13a) with $u(t, x)$. We then

approximate the mathematical expectation in (30) with an empirical mean, using sample paths of the Wiener process and the samples of $S^u(t)$ evaluated with the sample paths X_t . The second term in (30) depends only on the path of (13a) controlled by the suboptimal control u after time t . As indicated in (30), this term compensates for the residual between the suboptimal control $u(t, x)$ and the optimal control $u^*(t, x)$. In this context, the suboptimal control $u(t, x)$ acts as the proposal in terms of sampling techniques, such as the importance sampling [37]. Better proposals generally enable more efficient path sampling, resulting in smaller residuals in (30). Note that the path integral control (30) determines the control only at a specific time t , requiring the iterative computation of (30) at each time instant within the horizon $[0, T]$ [24].

5.2. Path integral particle filtering

As discussed in Section 4, computing the suboptimal controller (26) requires the particle approximation of the conditional distribution π_t^Y . We employ the particle filtering algorithm leveraging the path integral control to address the filtering, the path integral particle filtering (PIPF) algorithm proposed in [26,27], to obtain the particle approximation in this study. We briefly describe the algorithm, including slight modifications to integrate it with the proposed algorithm.

We focus on the filtering problem of the state of system (13a) based on the observation from observation equation (13b). The filtering problem for nonlinear systems is generally formulated as the problem of determining the conditional distribution of X_t given the observation history $\mathcal{Y}_t = \{Y_s\}_{0 \leq s \leq t}$. Below, we focus on the filtering of X_t given $\mathcal{Y}_t$ at a specific time t within the interval $[0, T]$. For this purpose, we assume that a control $\{u_s\}_{0 \leq s \leq t}$ is applied to the system (13) up to time t and is known.

The PIPF algorithm addresses the filtering problem via an optimal control problem. To this end, we consider the probability measure on the path of an SDE, which is a modification of (13a):

$$d\tilde{X}_s = \{f(\tilde{X}_s) + \sigma(\tilde{X}_s)u_s\} ds + \sigma(\tilde{X}_s) \left(\tilde{u}_s ds + \sqrt{\lambda} dW_s \right), \quad (31)$$

$$\tilde{X}_0 \sim \tilde{\pi}_0, \quad s \in [0, t],$$

where $\tilde{u}$ is a fictitious control for the optimal control problem related to the filtering problem. Below, we assume that the initial distributions π_0 in (13a) and $\tilde{\pi}_0$ in (31) satisfy $\pi_0 = \tilde{\pi}_0$ for simplicity. Then, we introduce the probability measure of the path $\{\tilde{X}_s\}_{0 \leq s \leq t}$ of the controlled system (31), denoted by $\tilde{\pi}_{[0:t]}$. Additionally, we can show that there exists the conditional distribution of the path $\{X_s\}_{0 \leq s \leq t}$ of Eq. (13a) given the observation history $\mathcal{Y}_t$, denoted by $\pi_{[0:t]}^Y$ (see [31, Theorem 2.1], for example). Note that $\tilde{\pi}_{[0:t]}$ and $\pi_{[0:t]}^Y$ are probability measures on the space of continuous functions over the interval $[0, t]$. Additionally, the marginal distributions $\tilde{\pi}_t$ and π_t^Y of $\tilde{\pi}_{[0:t]}$ and $\pi_{[0:t]}^Y$ provide the approximate and the true conditional distributions of X_t at time t , respectively.

The PIPF algorithm reduces the filtering problem to solving the minimization problem of the following functional [38]:

$$\mathbb{E}_{\tilde{\pi}_{[0:t]}} \left[\log \frac{d\tilde{\pi}_{[0:t]}}{d\pi_{[0:t]}^Y} \right] \quad (32)$$

where $\frac{d\tilde{\pi}_{[0:t]}}{d\pi_{[0:t]}^Y}$ is the Radon–Nikodym derivative of $\tilde{\pi}_{[0:t]}$ with respect to $\pi_{[0:t]}^Y$, and $\mathbb{E}_{\tilde{\pi}_{[0:t]}}$ is the mathematical expectation with respect to the probability distribution $\tilde{\pi}_{[0:t]}$. This is indeed the Kullback–Leibler (KL) divergence between the probability distribution of the controlled path of (31) and the conditional distribution $\pi_{[0:t]}^Y$. Furthermore, the Kallianpur–Striebel formula [31, Proposition 3.16] and the Girsanov formula yield an explicit expression of (32) as

$$\min_{\tilde{u}} \mathbb{E} \left[\int_0^t \left[\frac{1}{2\lambda} \|\tilde{u}_s\|^2 + \frac{1}{2\sigma_B^2} \|h(s, \tilde{X}_s)\|^2 \right] ds + \frac{1}{\sigma_B^2} \int_0^t Y_s^T dh(s, \tilde{X}_s) - \frac{1}{\sigma_B^2} Y_t^T h(t, \tilde{X}_t) \right], \quad (33)$$

where $\{Y_s\}_{0 \leq s \leq t}$ is the observation via the observation equation (13b) [38]. This implies that we can characterize the conditional distribution of X_t of (13a) by solving the optimal control problem of minimizing the functional (33) subject to the dynamics (31) with respect to $\{\tilde{u}_s\}_{0 \leq s \leq t}$. The resulting distribution $\tilde{\pi}_{[0:t]}$ of $\tilde{X}_t$ of (31) approximates the conditional distribution $\pi_{[0:t]}^Y$ in the sense of the KL divergence.

The PIPF algorithm provides a particle approximation of the conditional distribution π_t^Y as follows. First, we find a suboptimal control $\tilde{u}$ to minimize the functional (33) using a numerical technique, such as the iterative Linear Quadratic Gaussian (iLQG) method [39]. Then, using the suboptimal control, we compute K particles of the path $\{\tilde{X}_s^k\}_{0 \leq s \leq t}^{k=1, \dots, K}$ of (31). These particles approximate the conditional distribution π_t^Y at time t as required in (19) where the weights $\{w_f^k\}_{k=1, \dots, K}$ are given by

$$w_f^k = \frac{\exp(-S_f^{\tilde{u},k}(t))}{\sum_{k=1}^K \exp(-S_f^{\tilde{u},k}(t))} \quad (34)$$

and $S_f^{\tilde{u},k}$ is the value of the following function evaluated by using each sample path $\{\tilde{X}_s^k\}_{0 \leq s \leq t}$,

$$S_f^{\tilde{u}}(t) = \int_0^t \left[\frac{1}{2\lambda} \|\tilde{u}_s\|^2 + \frac{1}{2\sigma_B^2} \|h(s, \tilde{X}_s)\|^2 \right] ds, \quad (35)$$

$$+ \frac{1}{\sigma_B^2} \int_0^t Y_s^T dh(\tilde{X}_s) - \frac{1}{\sigma_B^2} Y_t^T h(\tilde{X}_t) + \frac{1}{\sqrt{\lambda}} \int_0^t \tilde{u}_s^T dW_s.$$

We employ the particle approximation through this PIPF algorithm in the computation of the controller (26).

5.3. Path integral control for partially observed systems

Given the particle approximation of the conditional distribution π_t^Y through the PIPF algorithm, it is only necessary to compute the suboptimal controller (26). The controller (26) is of the form of the conditional empirical mean of the optimal controllers for the fully observed optimal control, namely $-\sigma(x)^T \frac{\partial V^F}{\partial x}(t, x)^T$. The basic idea to compute the suboptimal controller (26) is that we use the path integral control algorithm, introduced in 5.1, to compute the components $-\sigma(X_t^k)^T \frac{\partial V^F}{\partial x}(t, X_t^k)^T$ in (26) and compute the conditional mean of them with the weights obtained from the particle filtering. Throughout this subsection, we assume that a particle approximation (19) of the conditional distribution is obtained using the PIPF algorithm described in Section 5.2.

We present details of the procedure to compute the suboptimal control (26). As in Theorem 1 in Section 5.1, the proposed framework employs a suboptimal control $u(t, x)$ as the proposal to obtain the optimal control $u^*(t, x)$ in formula (30). Given the particles $\{\tilde{X}_t^k\}_{k=1, \dots, K}$ at time t through the particle filtering, using (30), we compute

$$u^*(t, \tilde{X}_t^k) = u(t, \tilde{X}_t^k) + \lim_{\tau \rightarrow t} \frac{\mathbb{E}_{\tilde{X}_t^k} \left[\exp\left(-\frac{1}{\lambda} S_c^{u,k}(t)\right) \int_t^\tau \sqrt{\lambda} dW_s \right]}{(\tau - t) \mathbb{E}_{\tilde{X}_t^k} \left[\exp\left(-\frac{1}{\lambda} S_c^{u,k}(t)\right) \right]} \quad (36)$$

for each particle in $\{\tilde{X}_t^k\}_{k=1, \dots, K}$, where $S_c^{u,k}(t)$ is given by

$$S_c^{u,k}(t) = \int_t^T \left(\ell(X_s^{u,k}) + \frac{1}{2} \|u(s, X_s^{u,k})\|^2 \right) ds + \sqrt{\lambda} \int_t^T u(s, X_s^{u,k})^T dW_s + \Phi(X_T^{u,k}). \quad (37)$$

with $\{X_s^{u,k}\}_{t \leq s \leq T}^{k=1, \dots, K}$ being the solution to (13a) using the suboptimal controller $u(t, x)$, starting from $X_t^{u,k} = \tilde{X}_t^k$ at time t . The notation $\mathbb{E}_{\tilde{X}_t^k}^u$ denotes the mathematical expectation with respect to the path $\{X_s^{u,k}\}_{t \leq s \leq T}$ under the condition $X_t^{u,k} = \tilde{X}_t^k$. This component allows us

to compute the suboptimal controller (26) obtained in Proposition 1 as

$$u_t = \sum_{k=1}^K w_f^k u^*(t, \tilde{X}_t^k), \quad (38)$$

where $\{w_f^k\}_{k=1,\dots,K}$ are the weights in the particle filtering associated with the particles $\{\tilde{X}_t^k\}_{k=1,\dots,K}$. We apply the control u_t to the partially observed system (13).

Note that in the partially observed optimal control problems, it is generally challenging to specify the control due to the requirement to determine it from the conditional distribution, which involves a mapping from π_t^Y to $\mathbb{R}^m$. This difficulty arises because the partially observed control problem can be recast as the control problem for the conditional probability distribution, as seen from the discussion of the control problem of the Zakai equation at the end of Section 3. The controller (26) and (38) provide an approximate mapping through the particles $\{\tilde{X}_t^k\}_{k=1,\dots,K}$ and the associated weights $\{w_f^k\}_{k=1,\dots,K}$. This allows the controller to account for the state uncertainty through the conditional distribution in a practical manner. Consequently, the proposed controller (26) achieves a sufficient performance in the partially observed control problem of nonlinear systems, as demonstrated through numerical examples in Section 6.

5.4. Algorithm for partially observed control problems

We summarize our algorithm for the partially observed control problem, integrating the filtering and the control processes described in Sections 5.2 and 5.3, respectively.

We explain the implementation of the control (38). The proposed algorithm solves the PIPF and compute (38) iteratively at each time step $t \in [0, T]$. To compute the PIPF algorithm, we sample paths $\{\tilde{X}_s^k\}_{0 \leq s \leq t}^{k=1,\dots,K}$ through (31) and compute the associated weights $\{w_f^k\}_{k=1,\dots,K}$ in (34) as described in Section 5.2. Then, we compute the formula (36). To compute the second term in the right-hand side of (36), we sample the paths of the solution to (13a) on the interval $[t, T]$, starting from the values of particles $\{\tilde{X}_t^k\}_{k=1,\dots,K}$ obtained through the filtering. We sample L paths for every $\tilde{X}_t^k$, which is denoted by $\{X_s^{u,k,l}\}_{t \leq s \leq T}^{l=1,\dots,L}$ being the solution to (13a) with a given control $u(t, x)$ stating from $\tilde{X}_t^k$ (see also Fig. 1). We also evaluate the value of $S_c^{u,k}(t)$ for each sample $\{X_s^{u,k,l}\}_{t \leq s \leq T}$, denoted by $S_c^{u,k,l}$, through (37). The path integral formula (36) is approximated by

$$u^*(t, \tilde{X}_t^k) \approx u(t, \tilde{X}_t^k) + \sum_{l=1}^L \left(w_c^{k,l} \frac{\sqrt{\lambda} \Delta W_t^l}{\Delta t} \right), \quad (39)$$

where Δt is the time step size, $\{\Delta W_t^l\}_{l=1,\dots,L}$ is the sample of the increments of the Wiener process, $\Delta W_t^l = W_{t+\Delta t}^l - W_t^l$, and the weight $w_c^{k,l}$ is given by

$$w_c^{k,l} = \frac{\exp\left(-\frac{1}{\lambda} S_c^{u,k,l}\right)}{\sum_{l'=1}^L \exp\left(-\frac{1}{\lambda} S_c^{u,k,l'}\right)}. \quad (40)$$

With the weights $\{w_f^k\}_{k=1,\dots,K}$ obtained through the filtering and $u^*(t, \tilde{X}_t^k)$ obtained by (39), we evaluate (38). We iterate this procedure for each time step $t \in [0, T]$.

Algorithm 1 shows the entire procedure of the control algorithm for the partially observed control problems, assuming the discrete-time approximation. Lines 6–7 involve the procedure for the filtering, described in Section 5.2. Lines 8 to 11 involve the procedure for determining the control to the system, described in Section 5.3. In the numerical results presented below, we use the iLQG algorithm to design suboptimal controllers when using the path integral formula in the PIPF and (36), specifically in steps 6 and 8 of Algorithm 1.

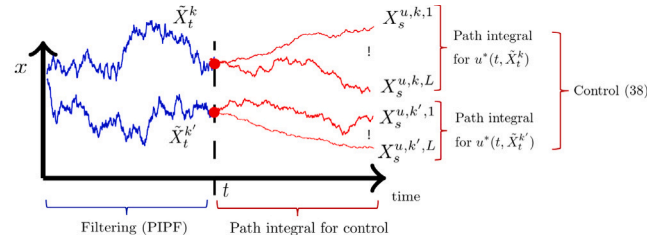


Fig. 1. Illustration of proposed algorithm.

Algorithm 1 Algorithm for Partially Observed Problem

- 1: Set time step size Δt , total number of time steps $N_T = \lceil T/\Delta t \rceil$, and numbers of particles K and L .
- 2: Sample initial particles $\{\tilde{X}_0^k\}_{k=1,\dots,K}$ from the initial distribution π_0 .
- 3: **for** $i = 0, 1, 2, \dots, N_T$ **do**
- 4: Set $t = t_i := i\Delta t$.
- 5: Measure the observation Y_{t_i} and store $\{Y_{t_j}\}_{j=0,\dots,i}$.
- 6: Find a suboptimal control to minimize (33) subject to (31) and compute discretized paths $\{\tilde{X}_{t_j}^k\}_{j=0,\dots,i}^{k=1,\dots,K}$ by the discretized equation of (31) with the suboptimal control.
- 7: Determine the conditional distribution (19) using weights $\{w_f^k\}_{k=1,\dots,K}$ from (34) and $S_f^{u,k}$ from (35).
- 8: Determine a suboptimal control $u(t, x)$ to compute proposals $u(t_i, \tilde{X}_t^k)$ in (36) for all the particles $\{\tilde{X}_t^k\}_{k=1,\dots,K}$.
- 9: Compute discretized paths $\{X_{t_j}^{u,k,l}\}_{j=i,\dots,N_T}^{l=1,\dots,L}$ through (13a) with $X_{t_i}^{u,k,l} = \tilde{X}_t^k$ for each k and evaluate $S_c^{u,k,l}$ by (37).
- 10: Compute controls $u^*(t_i, \tilde{X}_t^k)$ for particles $\{\tilde{X}_t^k\}_{k=1,\dots,K}$ using (39).
- 11: Determine u_{t_i} using (38).
- 12: Apply control u_{t_i} to the control system (13) and store u_{t_i} in the control history of $\{u_{t_j}\}_{j=0,\dots,i}$.
- 13: **end for**

6. Numerical example

We show numerical experiments of the proposed method to validate the effectiveness. We conducted two experiments of the unicycle model and the vertical take-off and vertical landing (VTOL) model. To evaluate the performance, we compare the proposed algorithm with a baseline method that uses the extended Kalman filter (EKF) and the iLQG controller (see [31, 8.1] for details on the EKF). The baseline method employs the iLQG controller as a feedback controller relying on the state estimates provided by the EKF.

(a) *unicycle model* We validate the proposed algorithm through a numerical example of a unicycle.

We examine the unicycle model, a standard benchmark of mobile robots, focusing on a parking problem under the partial observation. We consider the dynamical model of the unicycle with the 5-dimensional state variable given by $x = (x_1, x_2, x_3, x_4, x_5)^T \in \mathbb{R}^5$, where x_1 and x_2 represent the location of the vehicle on the plane, x_3 its orientation, x_4 the linear velocity, and x_5 the angular velocity (see Fig. 2). Assuming stochastic perturbations affect the input channel, we set $f(x)$ and $\sigma(x)$ of the state equation (13a) as

$$\begin{aligned} f(x) &= \begin{bmatrix} x_4 \cos(x_3) & x_4 \cos(x_3) & x_5 & 0 & 0 \end{bmatrix}^T, \\ \sigma(x) &= \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}^T. \end{aligned} \quad (41)$$

Accordingly, u_t and W_t are defined as a two-dimensional input variable and the two-dimensional standard Wiener process, respectively (see [40, Chapter 11] for the details of the unicycle modeling). Notably, the deterministic state equation of the unicycle, obtained by

Table 1
Expected Costs of the EKF-iLQG baseline and our proposed method.

Baseline	Proposed	Improved
5123.85	2677.59	47.74 %

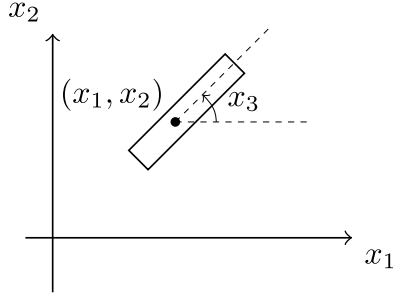


Fig. 2. Configuration of unicycle model.

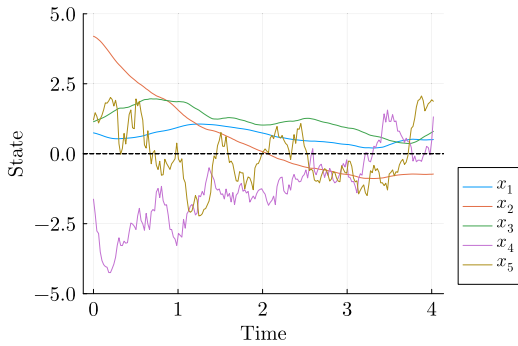


Fig. 3. Sample path of X_t via the proposed method.

neglecting the Wiener process W_t in (13a), results in an uncontrollable linear approximation of the system around the origin. Consequently, this requires considering the nonlinear model without resorting to the linear approximation. We assume that the function $h(x) = (x_1, x_2, x_3)^T$, representing the position and the angle, determines the observation equation (13b). Accordingly, B_t in (13b) is the three-dimensional standard Wiener process. The cost functional (14) of this example consists of the stage cost $\ell(x) = x^T Q x$ with $Q = 20I_5$ and the terminal cost $\Phi(x) = x^T Q_T x$ with $Q_T = 20I_5$. This cost design aims to steer the unicycle to the origin. In the numerical experiment, we choose $\sqrt{\lambda} = 1.0$, $\sigma_B = 0.2$, and used the terminal time of $T = 4$, the time step size of $\Delta t = 0.02$, and the numbers of samples $K = 100$ and $L = 1,000$. The probability distribution of the initial state is set to the Gaussian distribution with the mean $(2, 2, 0.5, 0, 0)^T$ and covariance matrix $2.0I_5$.

Table 1 shows the expected values of cost functional (14) obtained from Monte Carlo simulations of 50 independent experiments of the proposed and baseline methods. The results indicate that the proposed method performs better by 47.74% in terms of the expected cost values. Figs. 3 and 4 show example state trajectories obtained through the proposed and baseline methods, respectively. In the experiments, the performance of the baseline degraded when the EKF exhibited an error between the true and estimated state values, respectively. The proposed method did not often show such a degradation. This is because the proposed controller (26) accounts for the uncertainty of the system state through the particle approximation of the conditional distribution of the state, while the baseline method uses the approximate value of the conditional expectation of the state through the EKF.

(b) *Planar Vertical Take-off and Vertical Landing (VTVL) System* We validate our proposed method on a planar VTVL system. The state space is defined as $X_t \triangleq [x_G, y_G, \theta_G, v_{B_x}, v_{B_y}, \omega_{B_z}]^T$, where (x_G, y_G) are the positions of the mass center in the global coordinate, and (v_{B_x}, v_{B_y}) are

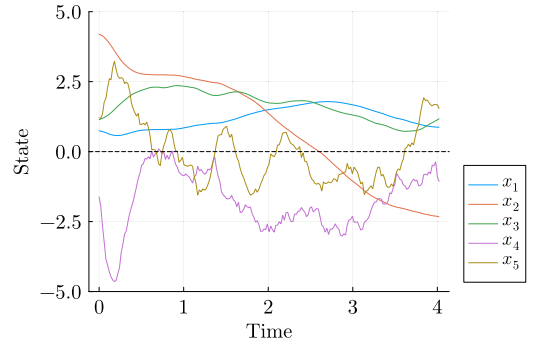


Fig. 4. Sample path of X_t via the baseline method.

Table 2
Expected Costs of the EKF-iLQG baseline and our proposed method.

Baseline	Proposed	Improved
19168.28	6859.16	64.22 %

the velocities in the body coordinate. (θ_G, ω_{B_z}) are the body angle in the global coordinate and its rotation speed in the body frame, respectively. The system has 3 control inputs, $u_t \triangleq [T_m, T_s, T_r]^T$, where T_m is the forward thrust, and T_s and T_r are 2 side thrusts. The functions of the dynamics (13a) are given by

$$f(x) = \begin{bmatrix} v_{B_x} \cos(\theta_G) - v_{B_y} \sin(\theta_G) \\ v_{B_x} \sin(\theta_G) + v_{B_y} \cos(\theta_G) \\ \omega_{B_z} \\ -g \sin(\theta_G) \\ -g \cos(\theta_G) \\ 0 \end{bmatrix}, \quad \sigma(x) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{m} & \frac{1}{m} \\ \frac{1}{m} & 0 & 0 \\ 0 & 0 & -\frac{l_r}{J_m} \end{bmatrix} \quad (42)$$

where g is the gravitational acceleration, m is the mass, J_m represents the moment of inertia, l_r is the width of the VTVL body. In this example, W_t is the three-dimensional standard Wiener process representing the noise. The function h of observation equation (13b) is given by $h(x) = [x_G, y_G, \theta_G]^T$.

We conducted a landing experiment for the VTVL system. The initial state is set to be

$$x_0 = [2.0, 2.0, 0.5, 1.0, 1.0, 0.1]^T. \quad (43)$$

The control problem is formulated to steer the state variable to the origin. The time window is set to be $[0, 3.75]$ with a discretization time step size 0.015. We used $K = 50$ particles for the filtering and $L = 200$ samples for the path integral control problem. The cost of the problem consists of a stage control energy cost, a stage quadratic state cost, $\ell(x) = \frac{1}{2} x^T Q x$, and a quadratic terminal cost $\Phi(x) = \frac{1}{2} x^T Q_T x$. Q is a zero matrix except for a quadratic penalty on the third dimension for θ_G , i.e., $Q(3, 3) = 150$, and $Q_T = 100.0I_6$. The noise intensity levels are $\sqrt{\lambda} = 1.0$, $\sigma_B = 1.0$. To estimate the expected costs (14), we conducted independent 50 experiments and used the Monte-Carlo empirical mean of the costs as the estimated expected cost. The cost statistics are shown in Table 2. Results show that our method can overcome the local optimality of the iLQG controller. Our method gained a 64.22% improvement in expected cost reduction. An exemplary state trajectory plot is shown in Fig. 5, where we compare the trajectory controlled by the proposed method and the baseline EKF-iLQG.

(c) *Algorithm implementation details.* Our proposed algorithm utilizes particles to represent the posterior distribution of the states conditioned on the observation history within the filtering component. For the control component, path integral control is applied to each particle individually, computed via forward sampling of the system dynamics under a proposal controller. Specifically, we select the iterative Linear

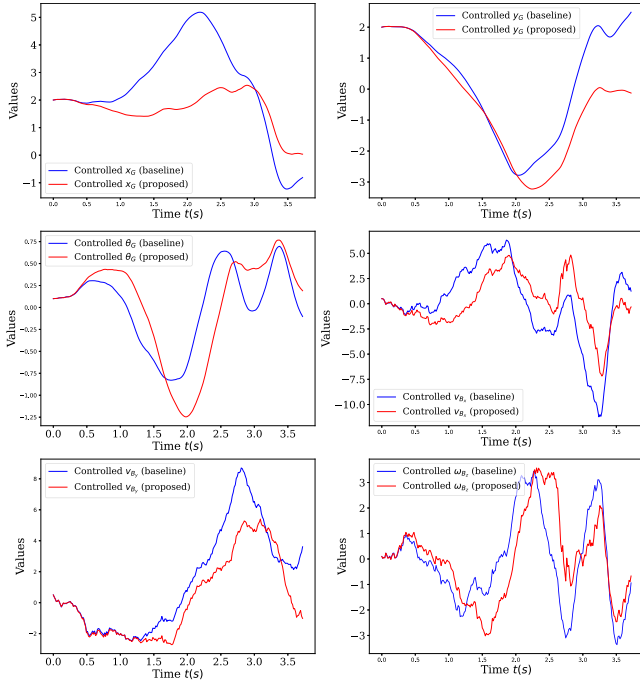


Fig. 5. Controlled state trajectories under the proposed and baseline EKF-iLQG strategies.

Quadratic Gaussian (iLQG) controller as the proposal controller, initialized from the current state of each particle. Consequently, the spatial complexity scales linearly with the product of the number of particles used in path integral filtering and the number of samples used in the path integral control.

Importantly, computational efficiency is significantly improved through parallelization. First, the particle filtering sampling step is parallelized on a GPU. Second, during the control stage, although path integral control is performed independently for each particle from different initial states, the sampling procedure across these particles shares a common proposal controller. The iLQG proposal controller is computed only once and then utilized by all particles. Additionally, the sampling process for all particles is efficiently parallelized on a GPU by concatenating their states into a single vector. Therefore, the time complexity of the optimal control computation becomes independent of the number of particles and aligns closely with the computational cost of performing path integral control given full state information.

In this work, our primary focus is algorithm development and demonstrating the advantages of particle-based state estimation compared to linearization-based local methods, such as the Extended Kalman Filter (EKF) and iterative Linear Quadratic Gaussian (iLQG), particularly for partially-observed control problems. We remark that implementing both particle filtering and the control component in a sliding-window manner can further improve runtime performance [26]. It is also worth mentioning that the sample complexity analysis for the proposed algorithm can be conducted based on that of the path integral control algorithm in the fully observed case [41]. Table 3 details the number of particles used in filtering, the number of samples employed in control, and the runtime in the unicycle the planar VTVL experiments. All the runtime data are collected by averaging 10 repeated experiments on a machine with an Intel Core i9-10920X processor and an Nvidia RTX 3090 graphics card.

7. Conclusion

This paper presents a control algorithm for the partially observed optimal control in continuous settings. This is achieved by developing

Table 3

Runtime performance of the examples.

Experiment	#Particles	#Samples	# Time steps	Runtime (s)
Unicycle (41)	100	1000	200	214.35
Planar VTVL (42)	50	200	250	729.93

a continuous analogue of the fully observable approximation of the POMDP. One of the main contributions of this study is the extension of the fully observable approximation, studied in discrete POMDPs, to the continuous-time and continuous-space settings for the first time. The continuous analogue of the approximation provides the suboptimal controller in the form of the conditional mean of the optimal controllers for a fully observed optimal control problem. It is noteworthy that the controller can be regarded as a generalization of the optimal controller in the linear-quadratic setting, designed through the separation principle. This indicates that our study contributes more than a straightforward extension of the fully observable approximation to the continuous setting. As a result of developing the approximation, we present the control algorithm which implements the suboptimal controller by leveraging the path integral control framework. The algorithm for the partially observed control problem effectively integrates the control and filtering methods through the path integral. As demonstrated through the numerical experiments, the proposed approximation method offers an efficient way to address partially observable optimal control problems, as the method outperforms the baseline in terms of performance and provides numerical results within a reasonable time frame of several minutes. Consequently, the proposed method is capable of handling the control problem of nonlinear systems.

Future works include the development of a more computationally efficient algorithm to enable real-time performance for an MPC algorithm for the partially observed optimal control, as developed for the fully observed scenarios [42,43]. In this direction, conducting a sample complexity analysis, as done for fully observed optimal control problems in [41], can be a promising direction for future work to ensure computational efficiency. Another possible direction is to apply the proposed method to more general class of control systems as in [44]. A further possible direction is the theoretical analysis of the proposed algorithm as mentioned in Remark 1.

CRedit authorship contribution statement

Kenta Hoshino: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hongzhe Yu:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Takashi Tanaka:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Yongxin Chen:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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